Smart Fisher Lanka is an AI-based decision support system designed for Sri Lankan fishermen

Project ID: 25-26J-177

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

....
Signature of the supervisor
(Prof. Pradeep Abeygunawardhena)

28/08/2025
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Date

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Signature of the co-supervisor
(Mr. Ashvinda Iddamalgoda)

2025/08/28
....
Date

Abstract

Sri Lanka's coastal fisheries sector plays an important role for thousands of small and medium-scale fishermen and their families. However, their fishing practices, including where they fish and how long they fish, are guided by experience, the weather, and trial-and-error. This can waste fuel, lead to poor predictability of catch, weaken bargaining power in the market, and reduce income, which this project aims to address through modern technologies applied to this sector.

In this project, we present a decision-support platform combining Artificial Intelligence (AI) and Internet of Things (IoT) technologies. For example, we will provide IoT devices including GPS loggers and catch loggers that can automatically log important spatial and temporal information about fishing trips, such as where and when fish are caught, and the catch volume. Fishermen will also be able to manually enter their catch log via a mobile application so that we can analyze both the automation and the user-generated data.

The data collected will be processed, cleaned and analyzed using AI-influenced models which can identify fish-rich zones, predict seasonal patterns and calculate the market prices. The AI system will provide recommendations and alerts that will be accessible through the mobile application to help fishermen make informed decisions about where and when to fish, while utilizing less fuel and time.

In addition, the initiative will create an online marketplace to connect wholesale buyers to fishermen directly. By harnessing real-time data, predictive analysis and transparent pricing, fishermen will enhance their income, the market efficiency and contribute to the long-term sustainability of Sri Lanka's coastal fisheries.

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List Of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
IoT	Internet of Things
GPS	Global Positioning System
CPUE	Catch Per Unit Effort
ML	Machine Learning
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
UX	User Experience
UI	User Interface
VMS	Vessel Monitoring System
API	Application Programming Interface
DBMS	Database Management System
SMS	Short Message Service

1. INTRODUCTION

1.1 Background

The coastal fisheries sector in Sri Lanka is economically important. When fishermen catch fish, it is still largely based on experience, hearsay, weather and their personal intuition. As a result, fishermen waste fuel by doing trial and error, they often do not know in advance what volume they will be able to land and are uncertain about the market or price for their fish, and receive poor prices for their fish harvests, with weak buyer support at their landing sites. Therefore, low-cost IoT hardware (GPS loggers, catch/event loggers, environmental monitors) and personal smartphones can be acquired to generate profits without having to pay, and modern AI methods can learn basic patterns from historical spatio-temporal data, allowing the fisherman to do their job more easily.

We propose an integrated AI-and-IoT data collection and decision-support platform for small (and medium-scale) fishermen:

- Boat Data Capture: GPS and `catch logger` automatically track the boat's location, time, and catch events, then the mobile app enables crews to manually log other species/weights.
- Cloud/edge analytics: Raw trip traces and catch records are cleaned and formatted, and AI identifies fish-rich zones and learns seasonal/daily patterns.
- Functional UX: The mobile app visualizes trip history, maps out trip destinations, and presents simple charts for crews to navigate the app and select the information they need.
- Market connectivity: Catch-logger data enables an online wholesale marketplace to surface live/anticipated landings with price analytics to improve seller leverage.
- Localized decision support: Notifications summarize near-term opportunities/risks. For example, "Avoid this zone today because the probability of catching fish is low. This is the recommended fuel-saving route. "The intended outcome is to reduce search time and fuel, stabilize incomes via better timing/targeting of grounds, and strengthen market outcomes through data-backed price insights.

1.2 Literature Review

Research and industry practice around “smart fisheries” spans sensing, prediction, compliance, and market intelligence. Key threads relevant to this work include:

- Vessel tracking & e-logbooks: Electronic logbooks and GPS/VMS have long been a standard for reconstructing effort and CPUE (catch per unit effort). Research continually shows that location-time-catch trajectories can convey seasonal migrations and productive habitats when anonymized and aggregated.
- Remote-sensing covariates: Sea-surface temperature, chlorophyll-a, fronts and upwelling indices are standard predictors of pelagic species presence with compositional feature sets developed from environmental layers routinely used in classical ML (Random Forest/Gradient Boosting) or sequence models (LSTM/Temporal CNN) to map "likelihood of catch".
- Spatio-temporal modeling: Grid-based heatmaps, kernel density estimates, and spatial point processes help uncover persistent hot-spots and shifting corridors. Recent work has created ensemble learners that incorporate time-lagged features that describe lead/lag effects between the environment and catch.
- Edge vs cloud trade-offs: Processing on the edge on mobile/boat devices minimizes upload costs and offline decision support, cloud pipelines are still preferred for model training and fleet-level pattern mining.
- Human-centered marine apps: UX research advocates low-friction logging (low touch points, defaults, voice dictation, etc.), offline charts, and simple, safety-first overlays. Trust is fostered when predictions are accompanied with confidence bands and a brief, plain language explanation.

Practical gaps in the literature for local contexts: (i) limited clean-labeled data at trip resolution; (ii) models that do not adapt to micro-regions/gear types; (iii) inadequate linkage between prediction and day-to-day operation; and (iv) limited link from at-sea data to price discovery at landing.

1.3 Research Gap

There are aspects of smart fisheries, but there is no complete end-to-end solution specific to Sri Lanka that firstly takes standardized high-level data from relatively inexpensive GPS and catch loggers, secondly converts it into local interpretable forecasts that can be used by fishermen at sea, and thirdly links it back to the market with price analysis related to expected landings.

The specific gaps that this project seeks to fill are:

1. Local predictive modeling: Most models are trained on large areas and the signals they extract are too coarse for small vessels. Our goal is to create micro-regional models that learn gear and species-specific patterns from logger data ,

2. environmental layers and visualize the results in easily confidence-rated heat maps and route recommendations.
3. Raw to Usable Data Pipeline: There are practical pipelines for cleaning, segmenting, and labeling raw GPS/catch streams into trip legs, event pulls, and CPUE features with minimal manual effort from the user. Our goal is to create a reproducible embedding and feature engineering pipeline suitable for intermittent connectivity.
4. Human in the Loop UX: Most existing tools stop at maps. We want to provide a mobile app with features like one-tap/manual catch access, history and charts, offline tile maps, and decision notifications that pair predictions with short “why this area” rationales that help build trust.
5. Market Intelligence Integration: Anglers rarely get transparent stock signals. Using catch-logger data, we prototype an online stock market and short-term price-trend forecast to advise when/where to fish.
6. Responsible Data Governance: Competitive sensitivity prevents adoption of fishing spots. We choose to share role-based, community dashboards that protect privacy, so leaders can control their data.
7. Seaton-Breeding Capability: Too many prototypes naively assume continuous signals across miles of steel. We bake in offline-first operation, delayed synchronization, and alternative edge inference for at-sea operations on commodity Android devices.

Novelty & Contribution:

- A logger-driven, localized decision support loop that can unify spate-temporal learning and fisherman-friendly rationales.
- Price analytics directly related to predicted/actual landings provide sellers with early notices of pending demand/supply patterns.
- A seamless platform of data capture → prediction → market connection that can support intermittent connectivity and privacy requirements of small-scale fleets.

1.4 Research Problem

Small- to medium-scale fishers in Sri Lanka rely on experience, intuition, and informal information for making operational decisions while at sea, often resulting in inefficiencies in finding fishing grounds, unanticipated catch volumes, and limited bargaining power at landing sites because of a lack of transparency around market information.

Despite a considerable international body of literature indicating the ways in which AI, IoT, and remote-sensing technologies can support "smart fisheries," these approaches are often coarse on a regional scale, are data-dependent, and rendered largely irrelevant for local fishers. The notable limitations pertain to:

- Lack of a standardized, inexpensive data collection mechanism associated specifically with local small vessels.
- The absence of localized AI predictions that fishers are able to act on and interpret in real time.
- Poor integration between decision support at-sea and market intelligence disempowers fishers at point of sale.
- Little or no support for functioning offline in low connectivity environments.

Consequently, the research problem presented is:

"How can an integrated AI- and IoT-based decision-support system be co-designed and co-developed to help Sri Lankan fishermen locate productive fishing areas, reduce operating inefficiencies, and enhance market outcomes though localized data driven insights and predictive analytics?"

2. OBJECTIVES

2.1 Main Objective

To design and develop an integrated AI and IoT–based data collection and decision-support system that assists fishermen in identifying potential fishing areas, recording and analyzing catch data, and predicting market trends through a user-friendly mobile application.

2.1Specific Objectives

1. Build IoT devices (GPS loggers, catch loggers) to automatically collect and report trip-level data such as location, time, and catch information.
2. Design a pipeline that facilitates our IoT raw data cleaning, processing, and transformation workflow into structured formats for AI fish zone prediction.
3. Apply AI methods to help anglers find potential fishing locations, analyze catch patterns, and enable localized decision-support notifications.
4. Build a mobile app that allows anglers to manually log catches during trips. Visualize previous catches on graphs/charts. View trip location and forecasts on maps.
5. Build an online wholesale marketplace for buyers to view and use catch log data, providing a real-time view of available fish stocks.

6. Analyze and predict potential market prices based on previous catch and sales data, and put fishermen in a better position to decide when to sell their catches.
7. Ensure offline-first data performance and prioritize data privacy; to highlight the usability of this system in low-connectivity areas and ensure that sensitive fishing data is protected.

3. METHODOLOGY

3.1 Requirement Analysis

A robust requirement analysis was performed to identify the relevant hardware and software components needed for the AI- and IoT-based fisheries decision-support system.

Hardware Components:

- GPS Logger Module – To capture the location, route and timestamp of fishing trips.
- Catch Logger Device – To log the catch events (species, quantity, weight).
- Microcontroller Board (e.g., Arduino / ESP32 / Raspberry Pi) – To integrate IoT sensors and have a local data hold.
- Communication Module (Wi-Fi/4G/Bluetooth) – To communicate between the IoT devices and a mobile application.
- Mobile Devices (Smartphones/Tablets) – Used by fishermen to run the mobile application and to log data and visualize.
- Cloud Server Hardware (Virtual / Hosted) – For central data storage and to deploy AI model.
- Power Supply / Battery Units – To give a power supply to IoT devices and to have them operate the entire duration of fishing trips.

Software Components:

- Mobile App (Android/iOS) - contains manual catch entry, history of trips, graphs, map displaying where catches were made.
- Embedded Firmware for IoT Devices - build modules for GPS and catching logging to collect and transmit raw data to the cloud.
- Cloud Database (firebase, AWS RDS, MongoDB) - storage of data with structure for synchronization for IoT data input.
- Data Processing Pipeline (Python/ETL scripts)- used to clean and transform raw data into datasets usable for vendors.
- AI/ML Frameworks (TensorFlow, PyTorch, Scikit-learn) - for predictive application of areas to fish and market price forecasting.

- Web App (Marketplace) - provide a connection between the fisherman and wholesale buyers with visibility into what is available including trends in price availability.
- APIs (REST/GraphQL)- communication between the mobile app, IoT devices to the cloud services.
- Security Tools (Auth, Role-based access, Encryption) - keep privacy and data from fisherman secure.

3.2 Feasibility Study

Strategic technical considerations:

- The IoT hardware (i.e., low-cost GPS and catch loggers) are available and inexpensive.
- Mobile applications can be built using cross-platform frameworks (e.g., React Native/Flutter) to support the application in iOS and Android.
- AI algorithms (for machine learning/deep learning) will have the capability for analytics to understand spatio-temporal fishing patterns, and to predict areas with high catch potential.
- Cloud and edge computing will provide storage and analytics functionalities with offline-first sync.

Economic feasibility:

- The initial costs will mostly be from sourcing IoT devices, paying for cloud services, and building the application.
- The long-term savings of reduced fuel wastage (and associated costs) and better decision-making in the marketplace will far outweigh the initial investment costs of buying the setup.
- The potential of some form of revenue from the online marketplace (for subscriptions/transaction fees).

Operational feasibility:

- Fishermen already own and use smartphones, and therefore their adoption of a mobile app will practical.
- There could be training sessions to assist fishermen in understanding how to use the mobile application.
- Overall, the system is set up to require a minimal number of manual inputs (as the majority of the data will be recorded automatically).

3.3 Implementation

The implementation will follow an iterative, modular development approach:

1. **IoT Module Development**
    - Configure and setup GPS loggers and catch loggers.
    - Assess data capture accuracy (time, location, catch event logging).
2. **Data Processing & Storage Layer**
    - Configure pipelines to clean raw IoT data and convert it into structured datasets.
    - Ensure storage of data in highly secure cloud database and with offline synchronization capability.
3. **AI & Analytics Engine**
    - Train AI models on past catch + environmental data.
    - Generate predictions for potential fishing grounds.
    - Develop market price prediction models using regression and time-series forecasting.
4. **Mobile Application**
    - Features: manual data entry, history, graphical dashboards, AI predicted maps.
    - Notifications: fuel saving recommendations, catch forecasts, market information.
    - Offline-first with synchronization when internet is available.
5. **Online Marketplace**
    - Provide a web-based marketplace for wholesale buyers.
    - Real-time access to fisherman’s catch availability combined with predicted pricing for upstream confidence.

3.3.1 System Diagram

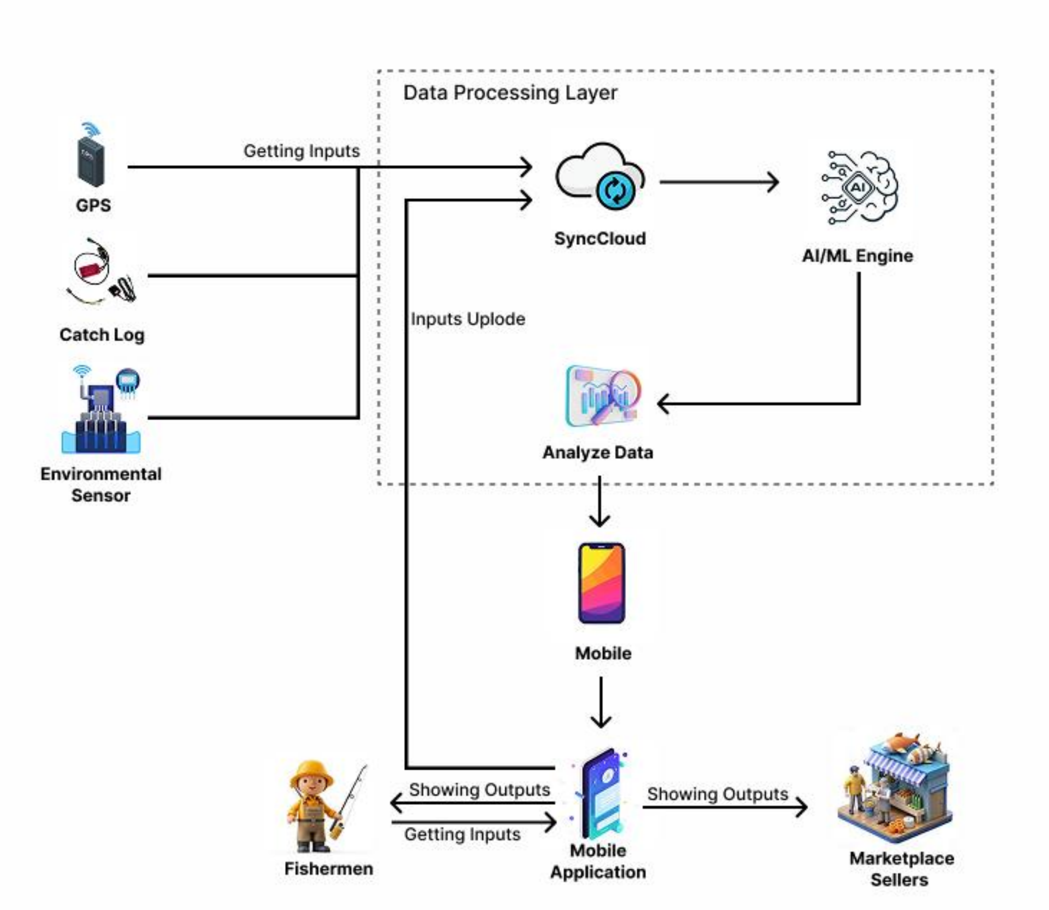

Figure 3.3.1.1.1 - System Diagram

4. PROJECT REQUIREMENT

4.1 Functional Requirements

The system must perform the following base functionalities:

1. IoT data collection - GPS loggers and catch loggers will automatically record the vessel's location, time, and catch to give the fishermen a location log of their catches.
2. Manual data capture - The fishermen should be able to manually log, via the mobile app, species, weight, and time of their catch.
3. Trip history & visualization - In the mobile app, fishermen should be able to view their prior trips, catch summaries, and graphs.
4. Mapping and prediction - On a map neighboring hot-spots, the mobile app should serve as a predictive mechanism for the hot-spot, e.g., via AI: its frequency and predictive capability can assist in attaining better catch.

5. Data processing & storage - Raw IoT data stored in a Hughes cloud database will be transformed into structured datasets for analysis purposes.
6. AI analytics - Predicting fish-rich areas and forecasting market prices.
7. Online marketplace - Wholesale buyers should be able to create view stock of what they can buy and the predicted prices to be expected.
8. Notifications & alerts - Decision-support alerts can be an additional service to fishermen. e.g., it may improve the environmental footprint such as fuel-saving routes, or directing fishermen to poor catch zones, etc.

4.2 Non-Functional Requirements

- The system has to meet the following quality attributes:
- Usability: The app interface will be easy and intuitive for fishermen who may have poor digital literacy.
- Scalability: The ability to integrate new IoT devices and support multiple fishing regions.
- Reliability: The system has to be reliably synchronized, even when in coastal environments with low connectivity.
- Performance: The predictions should be made in near real-time with minimal latency.
- Security & Privacy: The data of the fishermen will be encrypted and will have role-based authorization.
- Offline-first: The app must work with no continuous internet connectivity, and sync when it is able.

4.3 System Requirements

Hardware Requirements:

- GPS Logger and Catch Logger devices.
- Microcontroller (ESP32 / Arduino / Raspberry Pi).
- Smartphones (Android/iOS).
- Cloud Hosting Server containing AI and Database.

Software Requirements:

- Mobile Application (React Native / Flutter).
- IoT firmware for GPS and Catch loggers.
- Database (Firebase, AWS RDS, MongoDB).
- AI/ML libraries (TensorFlow, PyTorch, Scikit-learn).
- Web platform for marketplace (React/Angular + Node.js/Express).

4.4 User Requirements

Fisherman:

- Automatically or manually log catch data.
- View fishing history, maps, predictions, and alerts.
- Use mobile app offline and sync later.

Wholesale Buyers:

- View catch data available on the marketplace.
- Access short-term price predictions.
- Connect with fishermen on platform.

Admins / Researcher:

- Manage user roles and system data.
- Monitor AI predictions and analytics from the marketplace.
- Create reports and dashboards.

4.5. Work Breakdown Structure (WBS)

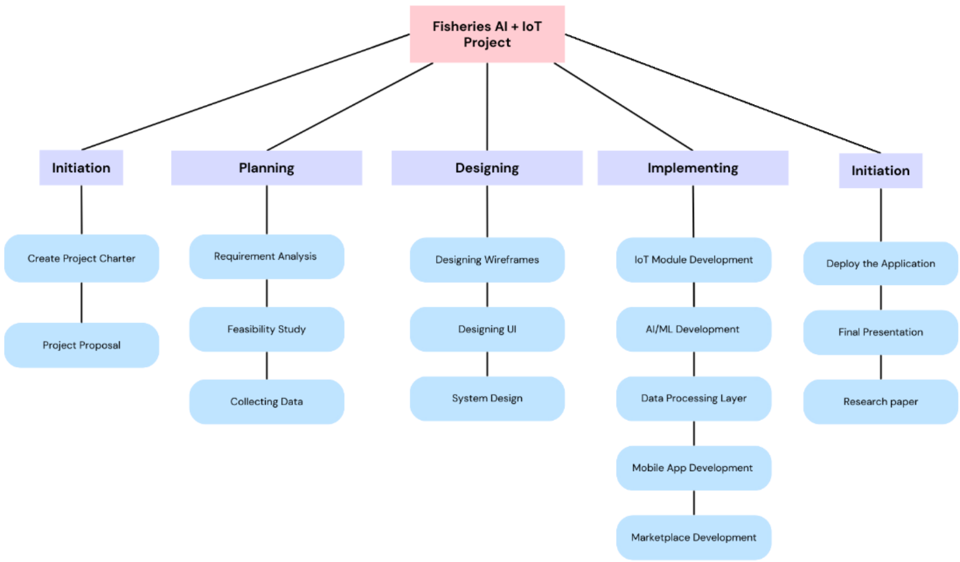

Figure 4.5.1.- Work Breakdown Structure

4.6. Gantt Chart

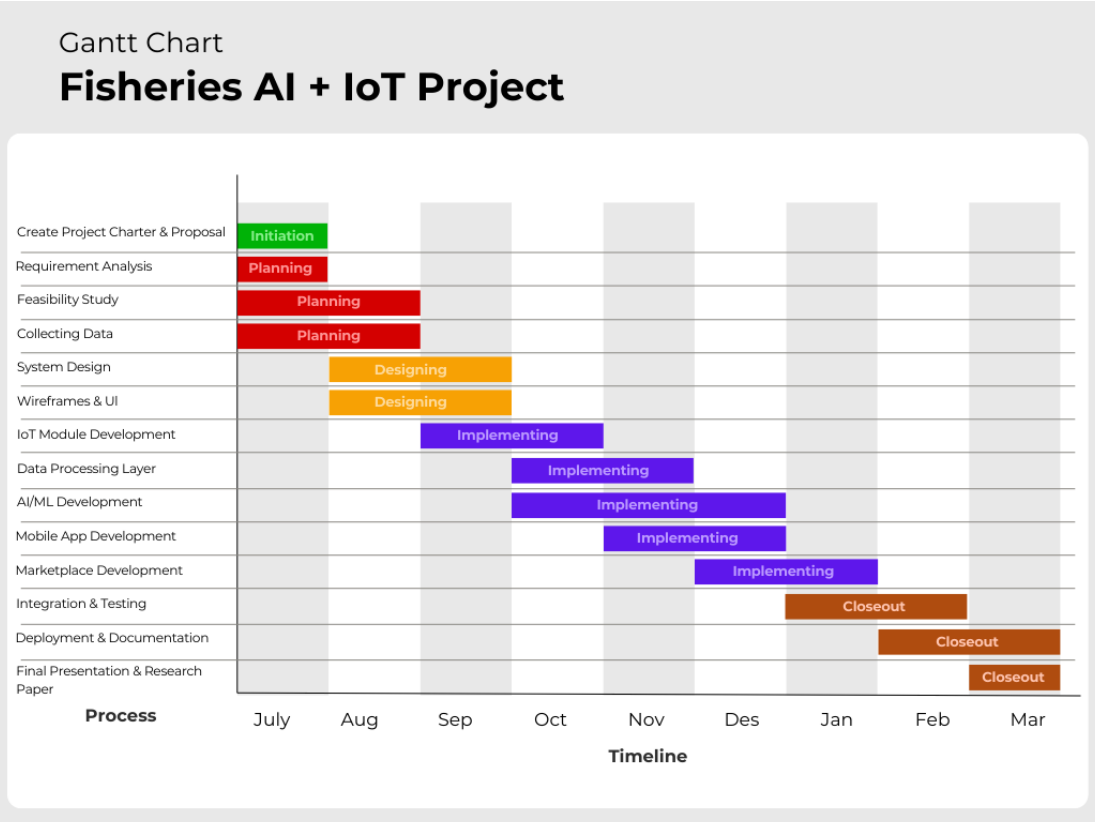

Figure 4.6.1. - Gantt Chart

5.DESCRIPTION OF PERSONNEL AND FACILITIES

5.1 Personnel

The proposed research requires a multidisciplinary team with expertise in fisheries, IoT hardware, AI/ML, mobile application development, and systems integration. The project involves the following roles:

- Project Supervisor (Academic Mentor): Provides overall guidance, ensures academic standards, and monitors progress.
- Research Team (Students): Responsible for requirement analysis, design, development, testing, and documentation.
- IoT Engineer: Designs and configures GPS loggers and catch loggers for the purpose of data capture.
- Software Developers: Build the mobile application, online marketplace, and backend database systems.
- AI/ML Specialist: Builds predictive models for fishing zone detection and market price predictions.

- UI/UX Designer: Ensures usability and accessibility of the mobile application.
- Domain Expert (Fisheries Consultant): Validates requirements to ensure system is aligned with fishermen's needs.

This composition ensures all elements of technical, domain, and usability aspects are fully covered.

5.2Facilities

The study will utilize the following facilities to conduct the research and develop and test the system:

- University Laboratory Facilities:
    - Computer labs with development tools, simulation software, and testing environments.
    - IoT hardware kits (Arduino, Raspberry Pi, ESP32 modules) for prototyping of GPS and catch loggers.3
    - Networking equipment and communications modules (Bluetooth, Wi-Fi and 4G modems) for IoT integration.
- Software Tools & Platforms:
    - Development environments: Visual Studio Code, Android Studio and Arduino IDE.
    - Cloud platforms: Firebase/AWS for storage, authentication and hosting.
    - AI/ML frameworks: TensorFlow, PyTorch, Scikit-learn for model development.
    - Database systems: MongoDB, MySQL/PostgreSQL for structured storage.
- Field Facilities:
    - Engagement with local fishing community for pilot testing.
    - Use of smartphones to test the mobile app in actual fishing trips.
    - Use of boats for testing GPS/catch logging devices and validating AI for predictive analysis.
- General University Facilities:
    - Access to libraries for journals and literature review.
    - Access to meeting rooms and facilities for review of progress.
    - Broadband internet and cloud services for remote collaboration and sharing data.

6.BUDGET ESTIMATION

Table 6.1. - Budget Estimation

Item	**Cost (LKR)**
GPS Logger Devices	35,000
Catch Logger Devices	50,000
Smartphones (for testing)	50,000
Cloud Hosting & Storage	30,000
Development Tools & Licenses	5,000
Personnel Cost (Stipend/Allowances)	80,000
Training & User Testing	15,000
Miscellaneous	10,000
Total Cost	275,000

References

(Matrix references, as you can add in IEEE/APA style according to your faculty requirement)

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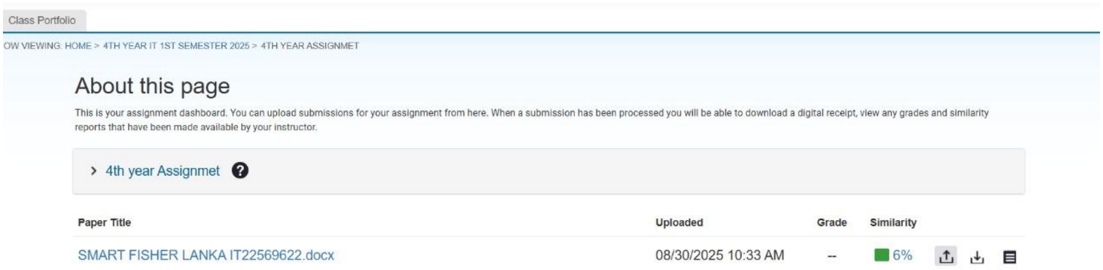
Appendices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